Law of cosines

1 $\triangle ABC$ consists of $\angle A$, $\angle B$ and $\angle C$ which appear opposite sides a, b and c respectively. If the measures of a, c and $\angle A$ are given, which rule should be used to find the size of $\angle C$, the law of sines or the law of cosines?

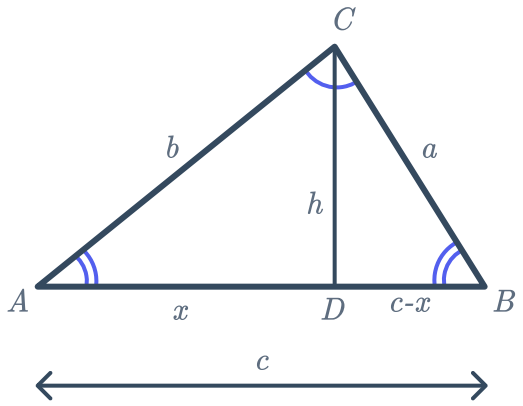
2 Consider the following triangle:

a Find an expression for a^2 by using Pythagorean Theorem in $\triangle BCD$.

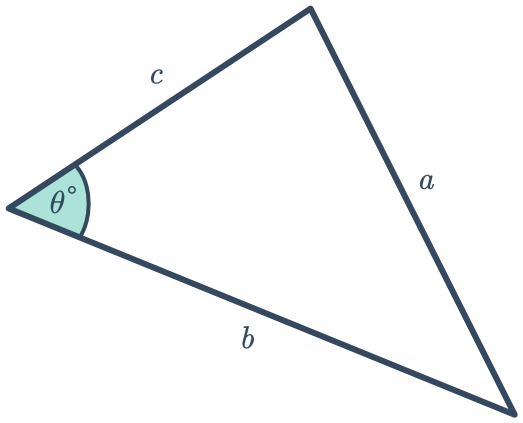
b Find an expression for h^2 by using Pythagorean Theorem in $\triangle ACD$.

c Find an expression for x in terms of $\cos A$.

d Substitute your expressions for h^2 and x into your expression for a^2 to prove the law of cosines.



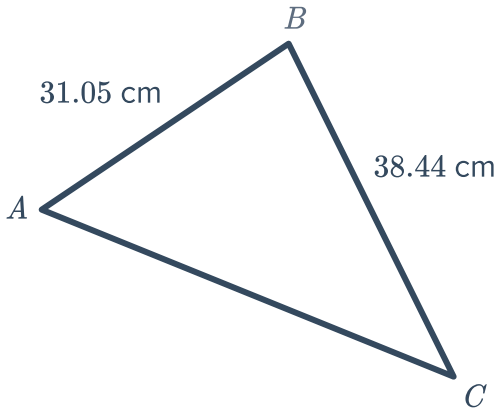
3 Write an expression for $\cos\theta$ using the law of cosines for the following triangle:



4 In $\triangle QUV$, $v = 6$, $q = 10$ and $u = 12$. Find the value of $\cos U$.

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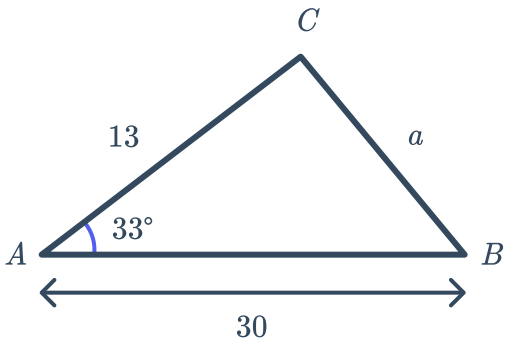
5 To use the law of cosines to find the length of $\overline{AC}$, which angle would need to be given?



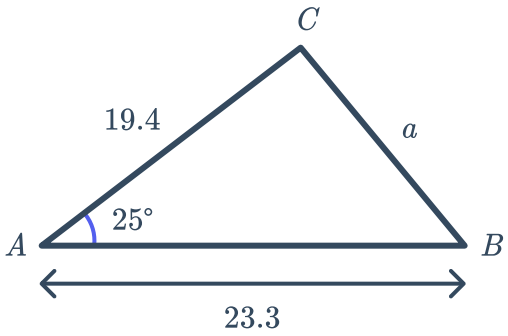
Unknown sides

6 Find the length of the missing side in each of the following triangles using the law of cosines. Round your answers to two decimal places.

a

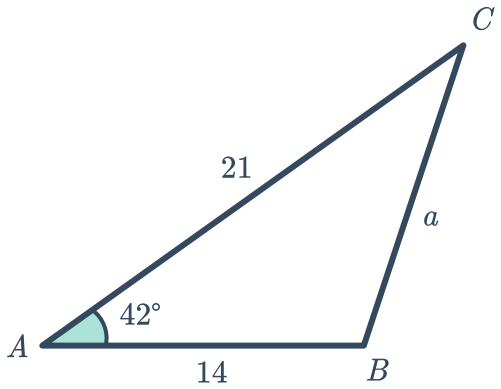


b

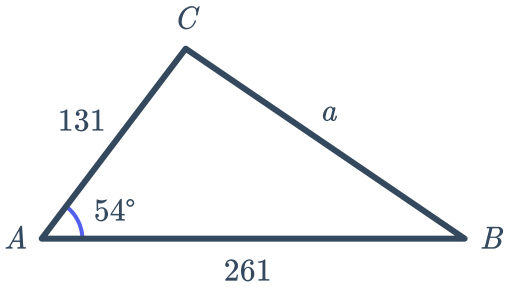


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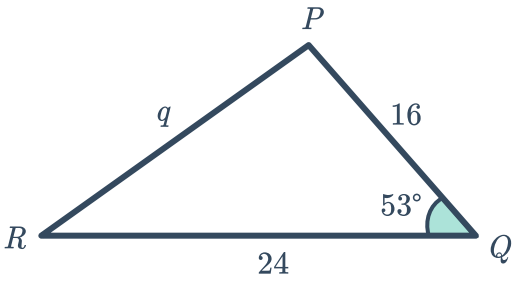
c



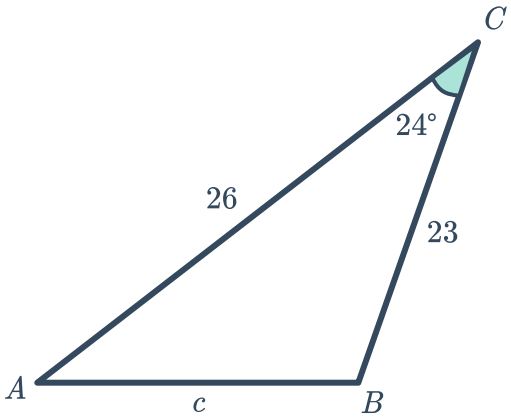
d



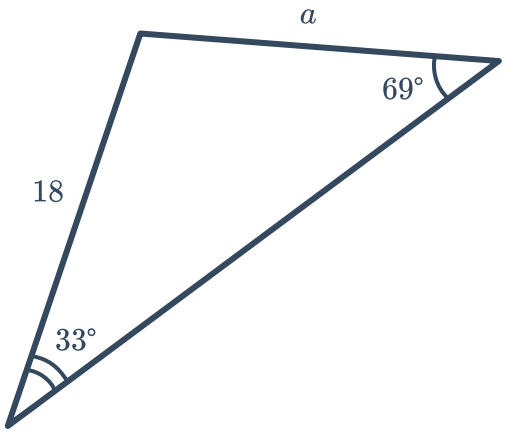
e



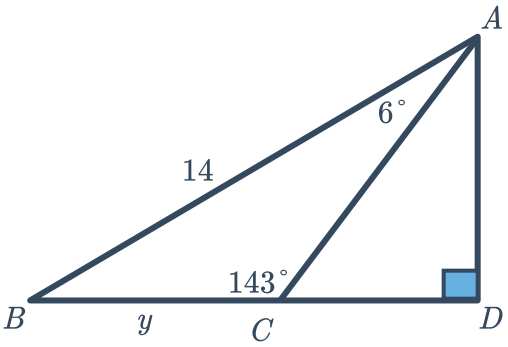
f



g



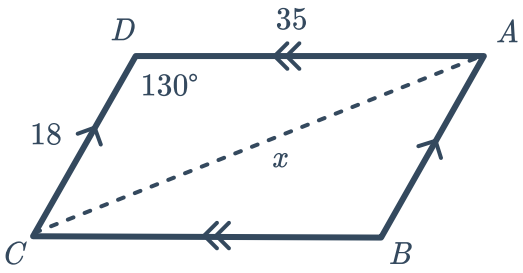
h



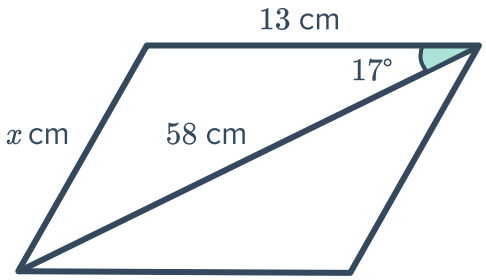
7 In $\triangle QUV$, $q = 5$, $u = 6$ and $\cos V = \frac{3}{5}$. Find the length of side v.

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8 Find the length of the diagonal, x, in parallelogram $ABCD$. Round your answer to two decimal places.

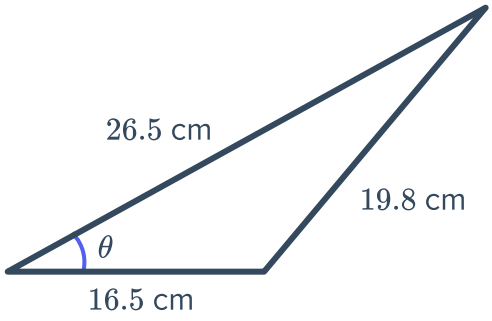


9 Consider the parallelogram in the given diagram that has a side of length $13\ \text{cm}$ and a diagonal of length $58\ \text{cm}$:
Find the value of x. Round your answer to one decimal place.



Unknown angles

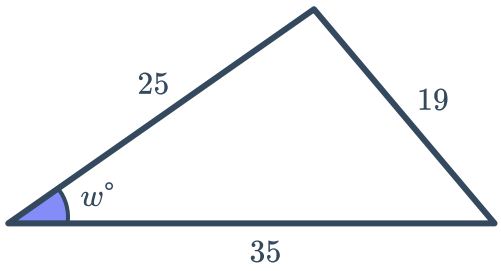
10 Find the measure of θ in the following triangle. Round your answer to the nearest hundredth of a degree.



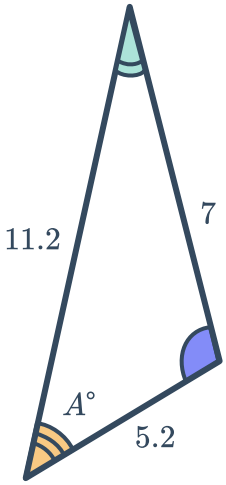
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11 For each of the following triangles, find the value of the variable in degrees. Round your answers to two decimal places.

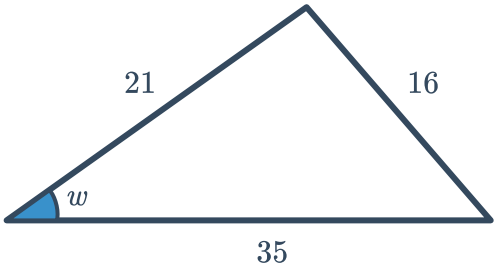
a



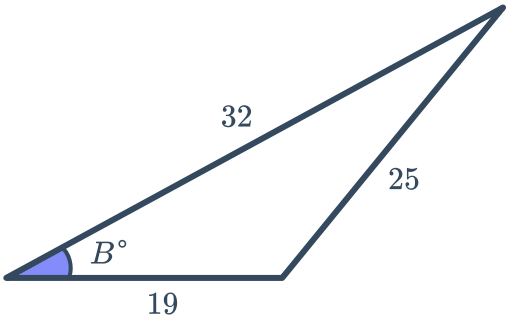
b



c



d



12 A teacher is writing exam questions for her math class. She draws a triangle, labels the vertices A, B and C and labels the opposite sides $a = 5$, $b = 8$ and $c = 15$ respectively. She wants to ask students to find the measure of $\angle A$. Explain the error with her question.

13 The sides of a triangle are in the ratio $4 : 5 : 8$. Find the measure of θ, the smallest angle in the triangle to the nearest degree.

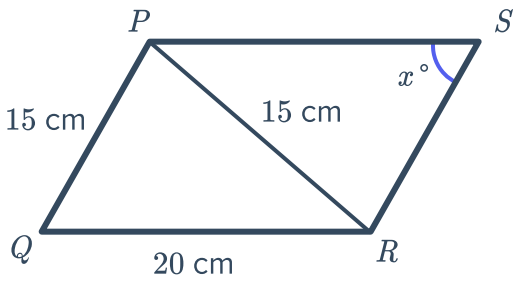
14 A triangle has sides of length 13 in, 15 in and 5 in. Find the measure of x, the largest angle in the triangle to the nearest degree.

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15 Consider the given parallelogram:

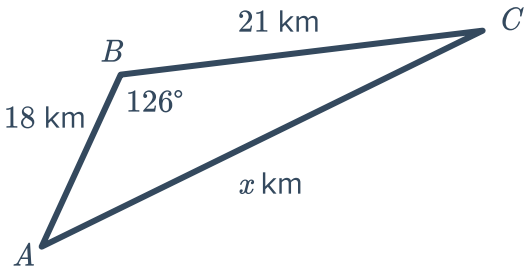
a Find the measure of x, to the nearest degree.

b Now, find the measure of $\angle SRQ$, to the nearest degree.

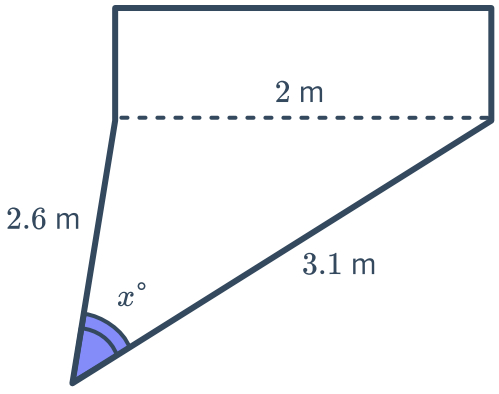


Applications

16 Mae went for a bike ride on Sunday morning from Point A to Point B, which was $18\ \text{km}$ long. She then took a 126° turn and rode from Point B to Point C, which was $21\ \text{km}$ long.
Find x, the distance in kilometers from her starting point to Point C to two decimal places.

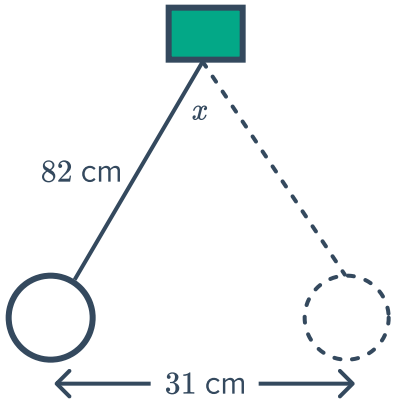


17 A goal has posts that are $2\ \text{m}$ apart. Buzz shoots for the goal when he is $2.6\ \text{m}$ from one post and $3.1\ \text{m}$ from the other post. Find the measure of the angle, x, in which he can score a goal. Round your answer to the nearest degree.

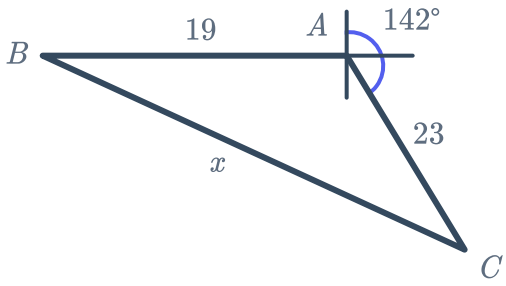


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18 A pendulum of length $82\ \text{cm}$ swings a horizontal distance of $31\ \text{cm}$.
Find the measure of the angle x of the pendulum's movement. Round your answer to the nearest degree.

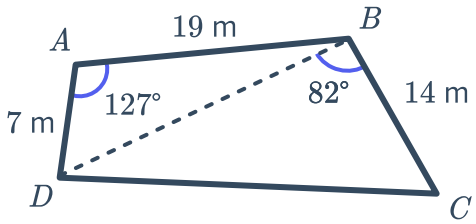


19 Point C has a bearing of 142° from Point A.
If Point B is $19\ \text{km}$ west of Point A, determine the distance, x, between Point B and Point C.



20 A garden, in the shape of a quadrilateral, is represented in the given diagram.
Find the following, rounding your answers to two decimal places:

- **a** The length of $\overline{BD}$.
- **b** The length of $\overline{CD}$.
- **c** The perimeter of the garden.



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21 Dave leaves town along a road on a bearing of 169° and travels $26\ \text{mi}$. Maria leaves the same town on another road with a bearing of 289° and travels $9\ \text{mi}$.
Calculate the distance between them to the nearest mile.

22 In a sailing boat race, teams must start at buoy A and navigate around buoys B and C before returning to buoy A to cross the line. The first leg of the race is $170.2\ \text{km}$ long, the second leg of the race is $150.9\ \text{km}$ long, and the angle between these legs is 111°.

a Find x, the distance of the third leg of the race. Round your answer to one decimal place.

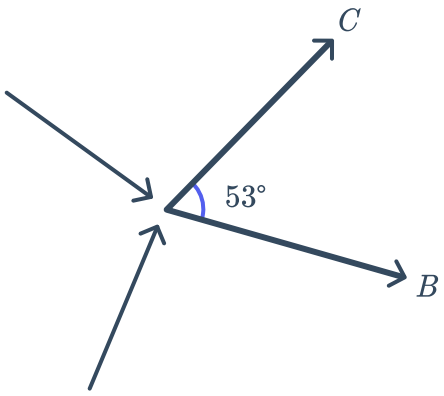
b Now, find the total length of the race. Round your answer to one decimal place.

23 After two meteoroids collide at point A, one starts traveling in the direction of point B, while the other starts traveling in the direction of point C, with an angle of measure 53° between the two directions. The meteoroid projected in the direction of B is moving at a speed of $7860\ \text{km/h}$, while the other is moving at a speed of $10\,170\ \text{km/h}$.

a Find the distance covered by the meteoroid traveling towards point B in the 29 minutes after the collision.

b Find the distance covered by the meteoroid traveling towards point C in the 29 minutes after the collision.

c Find the distance between the two meteoroids $29\ \text{min}$ after the collision. Round your answer to the nearest tenth of a kilometer.



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